

NCERT Class 12 Mathematics: Relations and Functions Formula Sheet

1. Types of Relations

A relation R in a set A is a subset of $A \times A$.

- **Empty Relation:** $R = \emptyset \subset A \times A$.
- **Universal Relation:** $R = A \times A$.
- **Reflexive Relation:** $(a, a) \in R$ for every $a \in A$.
- **Symmetric Relation:** $(a_1, a_2) \in R \implies (a_2, a_1) \in R$ for all $a_1, a_2 \in A$.
- **Transitive Relation:** $(a_1, a_2) \in R$ and $(a_2, a_3) \in R \implies (a_1, a_3) \in R$ for all $a_1, a_2, a_3 \in A$.
- **Equivalence Relation:** A relation R in a set A is an equivalence relation if R is reflexive, symmetric, and transitive.

2. Types of Functions

Let $f : X \rightarrow Y$ be a function.

- **Injective (One-to-One):** f is one-one if $f(x_1) = f(x_2) \implies x_1 = x_2$ for all $x_1, x_2 \in X$.
- **Surjective (Onto):** f is onto if for every $y \in Y$, there exists $x \in X$ such that $f(x) = y$. (Range = Codomain).
- **Bijjective:** f is bijective if it is both injective and surjective.

3. Composition of Functions

Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be two functions. The composition $g \circ f : A \rightarrow C$ is defined as:

$$(g \circ f)(x) = g(f(x)), \forall x \in A$$

4. Invertible Functions

A function $f : X \rightarrow Y$ is invertible if there exists a function $g : Y \rightarrow X$ such that:

$$g \circ f = I_X \quad \text{and} \quad f \circ g = I_Y$$

where I_X and I_Y are identity functions on sets X and Y respectively.

- A function f is invertible if and only if f is bijective.
- If f is invertible, then f^{-1} is unique.
- $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

5. Number of Relations and Functions

If $n(A) = p$ and $n(B) = q$:

- Total number of relations from A to $B = 2^{pq}$.
- Total number of functions from A to $B = q^p$.
- Number of one-one functions (if $q \geq p$) = $\frac{q!}{(q-p)!} = P(q, p)$.
- Number of onto functions (if $q \leq p$) = $\sum_{r=1}^q (-1)^{q-r} \binom{q}{r} r^p$.
- Number of bijections (if $p = q$) = $p!$.